

A Survey on Artificial Intelligence (AI) based Job Recommendation Systems

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Abstract—Job recommendation systems are AI-powered platforms that aim to provide job seekers with personalized job recommendations based on their skills, experience, and preferences. However, these systems face several challenges such as dealing with sparse data, maintaining data privacy and security, overcoming bias and discrimination, and ensuring transparency and interpretability. Recent techniques used in job recommendation systems include deep learning, reinforcement learning, and knowledge graphs. These techniques help to address the challenges such as dealing with sparse data, improving the accuracy of recommendations, and ensuring transparency as well as interpretability. The results of this study also helps to improve the job search experience for job seekers, reduces the time taken to find suitable job opportunities, and increase job satisfaction. Consequently, the job recommender architecture serves as a mediator. Accordingly, this work presents a detailed comprehensive survey of several filtering, machine learning and deep learning techniques that has revolutionised job recommendation system. Multiple applications and the associated challenges with existing job recommendation system are also discussed briefly.

Keywords: Job Recommendation System, Content Based Filtering, Knowledge Graph, Machine Learning

I. INTRODUCTION

Businesses affected by this pandemic disease slowly attempt to regain its lost strength as the entire world begins to stand again. Many employment firms have developed methods for offering the job board in order to serve the continuous cycle of recruiting process from the viewpoint of the job candidate. Here, a job seeker searches for and applies the positions that he thinks its relevant to him.

Joblessness is a significant issue that affects individuals, families, and entire communities. It can lead to financial instability and poverty, as well as mental health problems and a sense of hopelessness. The current economic climate has made the problem worse, with many businesses shutting down and job opportunities becoming scarce. Governments around the world are grappling with how to address this issue and provide support for those who are out of work. It's crucial that

a long-term solution is found to combat joblessness and help people get back on their feet.

Irrelevant job postings can be a major problem for job seekers, as they can waste a lot of time and energy searching for positions that they are not qualified for or do not meet their needs. By filtering out these irrelevant job postings and focusing on positions that are a good fit for the individual's qualifications and preferences, job seekers can save time and increase their chances of finding a suitable job. Additionally, this can also help to reduce the number of non-applicable job postings and make the job searching process more efficient for everyone. By offering tailored job recommendations to job seekers based on their abilities, geography, and other preferences, a recommender system can be an effective tool in the recruiting process. The user's qualifications and preferred jobs are gathered through this process, the recommender system can match them with job openings that are most relevant to them. This can save job seekers time and effort in their job search, as they will only be presented with positions that are a good fit for them. Additionally, by using a resume as input, the system can quickly and easily assess the user's qualifications and make recommendations without the need for additional forms or applications. Abbreviations used throughout the paper are listed below in *Table 1*.

The rest of this paper is organized as follows: details of different literature of job recommendation systems are illustrated in Section II. The detailed discussions of different types of job recommendation system are described in Section III. Comparisons of different methodologies of job recommendation system are summarized in Section IV. Challenges of building recommendation system briefly discussed in Section V. Section VI contains the overall discussion of recommendation system. Finally, Section VII concludes the work.

Table 1: Description of Abbreviations

RS	Recommender System
CF	Collaborative Filtering
CBF	Content Based Filtering
KB	Knowledge Based
SVD	Singular Value Decomposition
CV	Curriculum Vitae
CMAF	Competence Map
TF-IDF	Term Frequency-Inverse Document Frequency
SBERT	Sentence - BERT
KG	Knowledge Graph
NER	Named - Entity Recognition
MLM	Masked Language Modelling

II. LITERATURE SURVEY

[1] Using a combination of machine learning and data mining techniques to develop a job recommendation system can certainly help to bridge the gap between employees and employers. By leveraging the power of RESTful API and MongoDB, the system could potentially be more efficient and faster compared to traditional database management software like MySQL. It's great to hear that the authors acknowledged the presence of marginal errors in their system, but also mentioned that further research could help improve its accuracy. This shows their commitment to continuous improvement and suggests that their system has potential for further development and refinement.

[2] The research has made a valuable contribution to the field of resume recommendation systems. By using a machine learning approach and incorporating two different technologies (Content Based Recommendation using Cosine Similarity and k-Nearest Neighbors), they have developed a system that can accurately categorize resumes and recommend the most relevant ones based on the job description provided by recruiters. The two-phase approach (classification and recommendation) used by the researchers appears to be effective in capturing the insights and semantics of the resumes. The fact that their system achieved an accuracy of 78.53% is a strong indication of its effectiveness. This kind of accuracy can make a significant impact on the efficiency and effectiveness of the recruitment process.

[3] The research is significant in the field of job recommendation systems. By combining content-based and

collaborative filtering methods, the researchers were able to create a hybrid recommendation system that can leverage the similarities between applied-to jobs and new job opportunities. The results of the study showed that the hybrid recommendation system can prevent the need for extensive feature engineering or pre-clustering and can also effectively solve the cold-start problem. This is a great achievement, as it provides a more efficient and robust way to recommend jobs to users. The combination of these two methods can provide a more comprehensive and accurate recommendation, which can be of great benefit to job seekers and recruiters alike.

[4] They used the techniques of NLP, Linear Regression, Classification, Collaborative Filtering Techniques(CFR). By using a combination of NLP, linear regression, classification, and collaborative filtering techniques, they were able to create a job recommendation system that takes into consideration various important aspects of the applicant, such as their skills, experience, and qualifications. The use of NLP and collaborative filtering techniques allowed the researchers to create a recommendation system that can effectively analyze the relevance of a job opportunity to the applicant and provide a more accurate recommendation. The use of machine learning models was also a key component of the system, as it helped in making better decisions based on the data provided by the applicants and the job opportunities. Overall, the results of the study show that the proposed approach to job recommendation systems is effective and can provide a more comprehensive and accurate recommendation to job seekers. This can be of great benefit to both job seekers and recruiters, as it can help match job opportunities with the right candidates more efficiently and effectively.

The above information discusses various studies on job recommendation systems that use machine learning and data mining techniques. These systems can bridge the gap between employees and employers, improve the efficiency and effectiveness of the recruitment process, and benefit job seekers and recruiters alike. The studies used different approaches, such as content-based and collaborative filtering methods, NLP, linear regression, classification, and CFR techniques, and achieved high accuracy rates. The researchers acknowledged the presence of marginal errors and suggested further research to improve the systems' accuracy and efficiency. Overall, these studies show that job recommendation systems can be effective and provide comprehensive and accurate recommendations.

The literature survey is summarized below in *Table 2*.

Table 2 : Literature Survey Summary

Reference and Year	Methodology Considered / Merits	Demerits
[1], 2019	RestFul API, Android IDE	High Database Latency
[2], 2020	Machine Learning, Automation	Less Accuracy at 78.53%
[3], 2017	Content Based Filtering, Collaborative Filtering	Cold Start Problem
[4], 2020	Natural Language Processing, Regression, Classification	Combination of NLP and CFR

III. RECOMMENDATION SYSTEM

A system known as a recommendation engine employs machine learning algorithms to suggest the quality ones to a particular user or customer. It functions by spotting trends in consumer behavior data, which can be acquired through implicit or explicit methods.

Types of Recommendation Systems

There are mainly 4 types of recommendation systems as depicted in *Figure 1*.

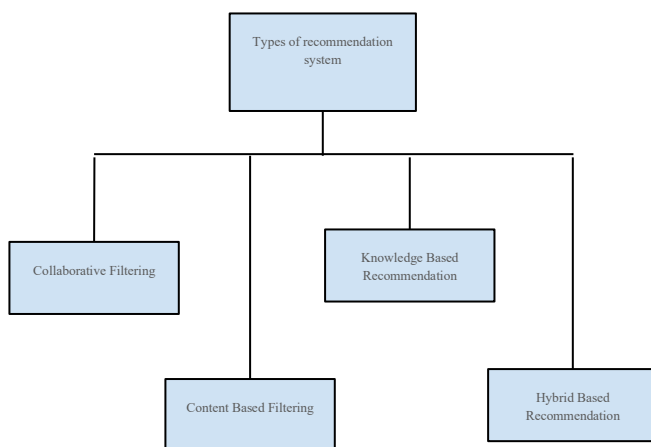


Figure 1 : Types of Recommendation Systems

A. Collaborative Filtering

This method of recommendation uses similarity measurements between users and/or items to suggest new items. The underlying assumption is that people with similar interests tend to have similar preferences.

B. Content Based Filtering

This approach to recommendation uses supervised machine learning techniques to classify items as interesting or uninteresting for a specific user. It is based on the idea that users tend to be interested in items that are similar to items they have liked or interacted with in the past.

C. Knowledge Based Recommendation System

This kind of RS aims to produce suggestions based on assumptions about the requirements and preferences of users [5].

This strategy aids job seekers in identifying the appropriate positions to apply for. These systems' recommendations are based on utilizing their extensive knowledge of jobs to identify the preferences of job seekers. In order to store the connections between job searchers' wants and employers' criteria, a knowledge base is needed.

D. Hybrid Based Recommendation System

These methods have been blended in order to get greater performance and overcome obstacles.

In order to overcome the drawbacks of other methods, collaborative filtering is typically combined with them [5]. The following *Table 3* lists various approaches suggested by Bruke [5][6] for integrating Collaborative Filtering, Content Based Filtering and Knowledge Based filtering into a hybrid Recommendation System.

Table 3 : Methods of Hybridization

Weighted	To create a single recommendation, the results of various recommendation systems are integrated.
Mixed	At the same time, recommendations from a variety of recommenders are displayed.
Switching	Depending on the circumstances, the algorithm alternates between different recommendation strategies.
Cascade	The recommendations made by one recommender are improved by another.
Feature combination	A single recommendation algorithm combines features from various sources of suggestion data.
Feature Augmentation	One technique's output is employed as an element of another's input.
Meta Level	One recommender's model is sent into another's as input.

IV. COMPARISON BETWEEN DIFFERENT METHODS FOR RECOMMENDATION SYSTEM

Filtering in recommendation systems involves selecting a subset of items to recommend to a user based on certain criteria. Some common filtering techniques used in recommendation systems are:

1. Collaborative Filtering: Compares the preferences of users who share similar interests to make recommendations for products based on similarities between individuals or objects. [7]

When creating individualized recommendations for consumers, collaborative filtering is a well-liked strategy. According to this theory, persons who have similar interests and preferences in the past will continue to have those same tastes and preferences in the future. Collaborative filtering can be divided into two primary categories: memory-based and model-based.

Memory-based Collaborative Filtering: In this approach, the algorithm calculates the similarity between users or between items based on their past interactions. The similarity can be calculated using various similarity metrics, such as Cosine similarity, Pearson correlation, or Euclidean distance. Once the similarity is computed, the algorithm can predict the preferences of a user for an item by aggregating the preferences of similar users.

Model-based Collaborative Filtering: In this approach, a model is trained to make recommendations. The model can be a matrix factorization model, such as Singular Value Decomposition (SVD), [8] or a neural network model. The matrix factorization model tries to factorize the user-item interaction matrix into latent features, which capture the underlying patterns in the data. The neural network model uses the interaction data as input and predicts the ratings or preferences of a user for an item.

Both memory-based and model-based collaborative filtering have their own advantages and disadvantages. Memory-based approaches are simple and easy to implement, but they can be slow and require a lot of memory to store the user-item interaction data. Model-based approaches can handle large amounts of data, but they are more complex and require more computational resources to train.

In conclusion, Collaborative Filtering is a powerful technique for making personalized recommendations and has been widely used in various applications, such as online retail, music streaming, and movie recommendations.

2. Content-Based Filtering: Recommends items based on their similarity to items the user has previously interacted with, by using the features or attributes of the items. [7]

Content-Based Filtering is another popular technique used in recommendation systems to make personalized recommendations to users. It operates on the idea that the recommendations should be based on the properties of the items, such as their genre, director, actors, or any other attributes.

The basic idea behind content-based filtering is to represent each item as a feature vector, where each feature is a characteristic of the item. For example, in a movie recommendation system, the features of a movie could be its genre, director, actors, and release year. The user's preferences for these features are then estimated based on their past interactions with items. Finally, the recommendation system generates recommendations by finding items with similar feature vectors to the items that the user has liked in the past.

There are two main steps in implementing content-based filtering:

Feature extraction: This step involves transforming each item into a feature vector, representing its properties. The feature vector can be represented as a binary vector, where each entry is 1 if the item has a particular feature and 0 otherwise, or as a continuous vector, where each entry is a numerical value that represents the strength of a feature.

User profile generation: This step involves estimating the user's preferences for the features of the items. This can be done by computing the weighted average of the features of the items that the user has liked in the past. The weights can be based on the ratings or preferences given by the user, or any other measure of user engagement with the item.

Once the user profile is generated, the recommendation system can generate recommendations by finding the items with the most similar feature vectors to the user's profile. The recommendations can be ranked based on the similarity score or any other measure of relevance.

In conclusion, content-based filtering is a powerful technique for making recommendations based on the properties of the items. It can be used to make recommendations in various applications, such as online retail, music streaming, and movie recommendations, where the items have well-defined properties or features. However, it can be limited by the quality of the features used to represent the items and may not be able to capture complex relationships between items and users.)

3. Hybrid Filtering: Combines the techniques of collaborative and content-based filtering to make recommendations based on both the preferences of users and the characteristics of items.

Hybrid Filtering is a combination of two or more recommendation techniques, such as Collaborative Filtering and Content-Based Filtering, used to generate recommendations for users. The goal of hybrid filtering is to leverage the strengths of each technique and overcome their limitations, resulting in more accurate and personalized recommendations. Hybrid filtering can be implemented in different ways, such as combining the results of multiple techniques, using one technique to address data sparsity and another to address the cold-start problem, and others.

4. Demographic Filtering: Recommends items based on the demographic information of users, such as age, gender, and location.

Demographic Filtering is a basic recommendation technique used in recommendation systems that makes recommendations based on the demographic characteristics of users, such as age, gender, location, and others. The recommendations are generated by finding the items that are popular among users with similar demographic characteristics and recommending them to the target user.

For example, in a movie recommendation system, demographic filtering might recommend action movies to male users and romantic comedies to female users based on the popularity of these genres among these groups.

Demographic Filtering is a simple and straightforward approach to recommendation systems and can be useful in situations where demographic information is readily

available and can be used to make meaningful recommendations. However, it does not consider the individual preferences of the user and may not generate accurate recommendations in all cases.

The choice of filtering technique depends on the type of data and the problem being solved. Additionally, different recommendation systems may use different combinations of filtering techniques to improve the accuracy and relevance of the recommendations.

Machine learning application

The creation of a job recommendation system can make use of a number of machine learning algorithms. The following are a few of the most popular algorithms:

1. Collaborative Filtering: This well-known algorithm for making recommendations is predicated on the notion that people with similar likes or preferences in the past are probably to continue to have those same preferences in the future. Methods like user-user collaborative filtering or item-item collaborative filtering can be used to implement collaborative filtering. In user-user collaborative filtering, the algorithm recognizes similar users and makes suggestions for products they enjoy. The algorithm finds comparable things and suggests items that are similar to the items that the user has previously liked in item-item collaborative filtering.

2. Matrix Factorization: This is an algorithm that is used to factorize a large sparse matrix into the product of two low-rank matrices. The idea behind matrix factorization is to discover latent features or hidden patterns in the data that can be used to make recommendations. For example, in a job recommendation system, the algorithm might discover latent features such as the type of job, the location, or the salary. These features can then be used to make recommendations to the user.

3. Content-Based Filtering: This is an algorithm that makes recommendations based on the attributes or features of the items being recommended. For example, in a job recommendation system, the algorithm might recommend jobs based on the user's education, experience, and skills. The algorithm analyzes the features of each job and recommends jobs that have similar features to the jobs that the user has liked in the past.

4. Decision Trees: A supervised learning technique that may be used to generate recommendations is called a decision tree. The programmer creates a model of decisions and their outcomes that resembles a tree and uses this model to make recommendations. For example, in a job recommendation system, the decision tree might consider factors such as the user's experience, education, and location, and make recommendations based on these factors.

5. Neural Networks: A form of machine learning algorithm known as a neural network is motivated by the structure and operation of the human brain. By discovering patterns in the data and creating predictions based on these patterns, neural

networks may be utilized to provide suggestions. For example, in a job recommendation system, the neural network might learn the relationships between different features of the job, such as the location, the salary, and the type of job, and make recommendations based on these relationships.

Each of these algorithms has its own strengths and weaknesses, and the best algorithm for a job recommendation system will depend on the specific requirements of the system and the data being used.

Frequently used Machine Learning algorithms

Collaborative filtering and content-based filtering are the two machine learning techniques that are most often employed in job recommendation systems.

Collaborative Filtering is a mainly executed machine learning algorithm in job recommendation systems because it is able to capture the relationships between users and jobs and make recommendations based on those relationships. The algorithm works by identifying similar users and jobs, and making recommendations based on what those similar users and jobs like. For example, if a user has applied for or shown interest in certain types of jobs in the past, the algorithm might recommend similar jobs to that user.

Collaborative filtering has the ability to manage enormous datasets and generate recommendations even when there is little information about a user accessible, which is one of its benefits. This makes it a suitable algorithm for job recommendation systems, where there may be a large number of users and jobs, but limited data available about each user's preferences.

Content-Based Filtering is also a frequently used machine learning algorithm in job recommendation systems. This algorithm makes recommendations based on the attributes or features of the jobs, rather than the relationships between users and jobs. For example, if a user is looking for a job in a certain location or with a certain salary range, the algorithm might recommend jobs that match those criteria.

The advantage of Content-Based Filtering is that it is able to make recommendations based solely on the job data, which can be particularly useful in situations where there is limited data available about the user. In these cases, the recommendations are based on the attributes of the jobs, rather than the user's past preferences.

A hybrid approach that includes collaborative filtering and content-based filtering is commonly utilized in job recommendation systems. This strategy can result in suggestions that are more precise and efficient since the Collaborative Filtering component can find individuals and tasks that are similar to one another and the Content-Depending Filtering component can tailor the recommendations based on the job attributes. A system of employment recommendations that is more sophisticated and personalized can be developed by combining these two methods.

Deep Learning Algorithms

Deep learning algorithms that can be used in a job recommendation system include:

1. Convolutional Neural Networks (CNNs): These are neural networks that are particularly well-suited for image classification and recognition tasks. In a job recommendation system, CNNs can be used to extract features from job descriptions, such as keywords, and use these features to make recommendations.[9]

2. Recurrent Neural Networks (RNNs): These are neural networks that are designed to handle sequential data, such as text. In a job recommendation system, RNNs can be used to analyze the job descriptions and predict which jobs a user is likely to be interested in based on their past preferences and the content of the job descriptions.[10]

3. Autoencoders: These are unsupervised deep learning algorithms that can be used to learn compact representations of data. In a job recommendation system, autoencoders can be used to learn compact representations of job descriptions, which can then be used to make recommendations based on the similarity of the job descriptions.

4. Generative Adversarial Networks (GANs): These are deep learning algorithms that are used to generate new data that is similar to the data in a training set. In a job recommendation system, GANs can be used to generate new job descriptions that are similar to the job descriptions that the user has shown an interest in and recommend these new job descriptions to the user.[11]

Each of these deep learning algorithms has its own strengths and weaknesses as compared in *Table 4*, and the best algorithm for a job recommendation system will depend on the specific requirements of the system and the data being used. In general, deep learning algorithms can provide more sophisticated and personalized recommendations compared to traditional machine learning algorithms, but they also require more computational resources and can be more difficult to train and fine-tune.

Table 4 : Comparison between ML and DL Algorithms

	Machine Learning Algorithms	Deep Learning Algorithms
Training Complexity	Lower	Higher
Computational Resources	Lower	Higher
Personalization	Good	Better
Scalability	Good	Good
Adaptability	Good	Better
Ability to Handle Complex Data	Limited	Good

Training Complexity: Machine learning algorithms are generally easier to train compared to deep learning

algorithms, as deep learning algorithms often require more data and computational resources to train effectively.

Computational Resources: Deep learning algorithms typically require more computational resources compared to machine learning algorithms, as they have more parameters and require more data to train effectively.

Personalization: Deep learning algorithms tend to provide more personalized recommendations compared to machine learning algorithms, as they are able to handle more complex and diverse data and learn more sophisticated relationships between users and jobs.

Scalability: Both machine learning algorithms and deep learning algorithms can be scaled to handle large datasets, but deep learning algorithms may require more computational resources to scale effectively.

Adaptability: Deep learning algorithms tend to be more adaptable compared to machine learning algorithms, as they are able to learn new relationships and patterns in the data more effectively.

Ability to Handle Complex Data: Deep learning algorithms are better suited to handle complex and diverse data compared to machine learning algorithms, as they can learn more sophisticated relationships between users and jobs. However, this also means that deep learning algorithms may require more data and computational resources to train effectively.

V. CHALLENGES IN GENERAL FOR ANY RECOMMENDATION SYSTEM

Inability to capture changes in user behavior-

The behaviors and purchasing habits of consumers are always evolving. They aren't constant. It's a constant challenge to stay on top of these advancements.

A strong recommendation engine will be able to recognize shifts in consumer preferences and behavior and continually auto-train in real time to provide relevant suggestions.

A way to implement a feedback loop is to provide users with the ability to rate or provide feedback on recommended items. The system can then use this feedback to refine future recommendations, taking into account the user's preferences and feedback. The system can also use machine learning algorithms to analyze the feedback data and adjust the recommendation model accordingly.

Lack of data analytics capability-

High computational complexity may be required for recommendation engines.

If the input data are less valuable or dependable, the model's output will be less beneficial. Therefore, before spending money on recommendation engines, it should be made sure that application can handle the complicated data analytics requirements.

One can leverage third-party analytics tools, hire data analysts, incorporate machine learning algorithms, and utilize user feedback to improve the system's data analysis and recommendation accuracy.

The complex onboarding process-

These may also be the result of a variety of other factors, including a lack of comprehension of recommendation algorithms, substandard website UX, and inadequate business domain knowledge. They may utilize recommendation engines if engaged and managed the correct personnel who are knowledgeable about both technology and the detailed operations of your application.

To overcome the complexity of the onboarding process for a job recommendation system, one can simplify the registration process, provide clear instructions and guidance, incorporate social media logins, and offer personalized onboarding experiences based on the user's profile and preferences.

Concerns privacy-

The more details the algorithm has about the user, the more accurate its suggestions will be. Many consumers are hesitant to divulge personal information, however, in light of recent high-profile incidents of consumer data leaks. However, without this client data, the recommendation engine would not function effectively. Therefore, building consumer and business trust is essential.

It is essential to implement strong data protection policies, ensure compliance with relevant data privacy regulations, provide transparency on data usage, offer opt-out options, and obtain user consent before collecting and using their data. These measures can help to build user trust and confidence in the system while also ensuring the protection of their personal information.

Multiple choices-

It could take a while to compare different solutions since they have to take into account their case studies, technology, integration with your current business strategy, and other considerations.

One can implement personalized filtering and sorting options that align with user preferences, utilize machine learning algorithms to understand user behavior and provide relevant job recommendations, provide clear and concise job descriptions that accurately reflect the job requirements, and offer diverse job recommendations based on user feedback to enhance the user experience.

[12] In order to complement the efforts of both job seekers and recruiters, this article suggests a reciprocal recommendation system built on two-way communication. Job searchers may locate the best-fit positions to match their resumes, and with the help of their job advertisements, recruiters can identify the top candidates for each position. In this, it shows how machine learning may be used to tackle problems with the natural language processing of text content and similarity scores based on job offers.

To determine similarity for bi-directional matching, explicit and implicit job information from two sides (recruiters and job seekers) has been integrated. The suggested method is assessed using a dataset of resumes and job offers. Utilizing decision support tools, created recommendations' effectiveness is assessed.

The obtained findings show that the suggested approach is capable of improving prediction accuracy as well as resolving the issue of bi-directional recommendations.

[13] In order to incorporate rate as a feature into job seeker model with other features, this study focuses on creating practical embeddings of textual information from job postings and resumes. exemplifies a profession where domain-specific challenges are posed by noisy data from parsed resumes, the diversity of the different data sources, and cross- and multilingualism.

uses a huge, accurate, and high-quality dataset of over 270,000 resumé-vacancy pairings identified by our staffing experts to optimize Siamese Sentence- BERT (SBERT) model, which names consultant BERT, in order to overcome these issues. Which shows that our improved model outperforms unsupervised and supervised baselines that rely on TF-IDF-weighted feature vectors and BERT embeddings by a wide margin. This method also works well with cross-lingual and multilingual textual material. Set of training data to optimize the embedding models.

[14] Studying the distributions of prediction scores across the two labels (positive and negative matches) and zooming in on another desirable characteristic of our subspace the various approaches in further detail.

The two key responsibilities in the hiring process are recruiter and candidate. Despite the fact that there are many available positions and few competent individuals to fill them, the goal is to only provide client profiles that match their needs. The challenges of today include the demand for adaptive data for the specific staffing responsibilities of BERT as well as the regular evolution of experiences in Curriculum Vitae (CV). This study provides a system for classifying candidates based on competence-based keywords. There are four phases. First, use the Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer to calculate the score of matching between a competence term and a corpus of CVs. Second, depending on the function and specialty of a competency keyword, we compute a weighted average utilizing the weighted average approach.

Third, use the structured Competence Map (CMAP) to generate a Knowledge Graph (KG), which can classify the aggregate and bidirectional connection linkages. Finally, using Named-Entity Recognition (NER) and Masked Language Modeling (MLM) features of BERT to more precisely identify tokens from the client's input enquiry.

VI. DISCUSSION

With the help of all the papers it is found out various techniques and methods about the recommendation system. Using NLP and NER for extracting information from resumes is a common approach for streamlining the job application process. By automating this process, it can make it easier for users to input their information, which can save them time and effort. Additionally, by using BERT for NER, it can ensure that the extracted information is highly accurate and relevant. However, it is important to keep in mind that some resumes might be formatted differently, and this could impact the accuracy of the information extraction process. It may be

necessary to develop techniques to handle these differences, such as using multiple NER models or training their own custom NER model on a large corpus of resumes.

VII. CONCLUSION

In conclusion, job recommendation systems have the potential to greatly benefit job seekers by providing personalised and accurate job recommendations.

The objective of this study is to develop a job recommendation system that addresses these challenges and provides accurate and unbiased job recommendations to job seekers. The hybrid approach that combines collaborative filtering and content-based filtering, and incorporate techniques such as differential privacy, fair representation learning, and model interpretability methods will help in achieving higher accuracy. As it combines multiple techniques. It uses various methods such as weighted, mixed, cascading, feature augmentation etc to give more accurate results.

The effectiveness of the system will be evaluated through user studies and comparisons with existing job recommendation systems. The results of this study can help improve the job search experience for job seekers, reduce the time it takes to find suitable job opportunities, and increase job satisfaction.

Overall, this study provides a comprehensive survey of the various techniques used in job recommendation systems and highlights the challenges associated with existing systems. This work can serve as a valuable resource for researchers and practitioners in the field of job recommendation systems.

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